II Year B.Tech. CE – II Semester

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(23A01402P) CONCRETE TECHNOLOGY LABORATORY

Course Objectives: To test basic properties of ingredients of concrete fresh and hardened concrete properties

Course Outcomes:

CO1 Outline importance of testing cement and its properties
CO2 Assess different properties of Aggregates
CO3 Assess fresh concrete properties and their relevance to hardened concrete
CO4 Assess hardened concrete properties

Detailed Syllabus:

1.Tests on Cement

Normal Consistency and Fineness of cement.

Initial setting time and Final setting time of cement.

Specific gravity and soundness of cement.

Compressive strength of cement.

2.Tests on Fine Aggregates

Grading and fineness modulus of Fine aggregate by sieve analysis.

Specific gravity of fine aggregate

Water absorption and Bulking of sand.

3.Tests on Coarse Aggregates

Grading of Coarse aggregate by sieve analysis.

Specific gravity of coarse aggregate

Water absorption of Coarse aggregates

4.Tests on fresh Concrete

Workability of concrete by compaction factor method

Workability of concrete by slump test

Workability of concrete by Vee-bee test.

5.Tests on Hardened Concrete

Compressive strength of cement concrete and Modulus of rupture

Young’s Modulus and Poisson’s Ratio

Split tensile strength of concrete.

Non-Destructive testing on concrete (for demonstration)